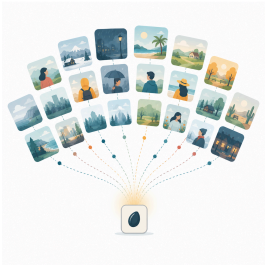
C1 Asset Construction GAP — 19%

1. **Assets as conditional input**
   MAGE · DriveDreamer
2. **World model generates assets**
   WonderWorld · Free4D
3. **Unified editable platform**
   WorldGen · GigaWorld-0

Explicit-3D/4D · Diffusion

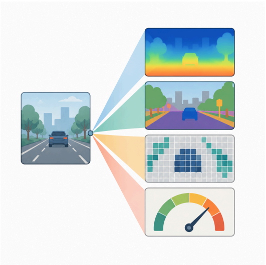
C7 Diversity — 23%



1. **Scale-driven**
   Cosmos · GenAD · UniSim
2. **Composition-driven**
   RoboDreamer · DriveDreamer-2
3. **Parametric-controllable**
   GigaWorld-0

Diffusion · Latent-dyn

C6 State Feedback GAP — 22.5%



1. **Multi-task auxiliary outputs**
   Tesseract · MUVO · DiST-4D
2. **Derived from physics**
   OrbiSim · Kinema4D
3. **Latent-space representation**
   V-JEPA2 · Cosmos Policy

Occupancy · Diffusion (truth-only 8%)

C3 Interaction STRENGTH — 40%

1. **Camera-level control**
   GameNGen · Matrix-Game
2. **Object-level interaction**
   ActWorld · WorldCraft
3. **Multi-agent / algo-in-loop**
   DIAMOND · V-JEPA2-AC

Diffusion · Autoregressive

World Model *as* Simulator

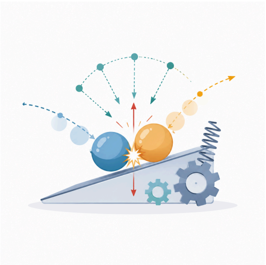
Eight capabilities of a simulator, mapped across generative world models

action / camera / language → **[state + dynamics]** → RGB + structured feedback

■ strength ■ structural gap ■ neutral

coverage % = share of 200-paper corpus

C2 Physics Engine GAP — 17%



1. **Hard physics core**
   OrbiSim · PIN-WM · PWTF
2. **Soft physics constraints**
   GEM-4D · PEVA · FlowDreamer
3. **Physics evaluation only**
   What-If World · PhysBench

Diffusion · Latent-dyn

C4 Controllability STRENGTH — 62.5%

1. **Explicit control**
   DisCo · Prisma-World
2. **Language / multimodal**
   Pandora · DrivingGPT · UniPi
3. **Latent action**
   Genie · AdaWorld · DiLA

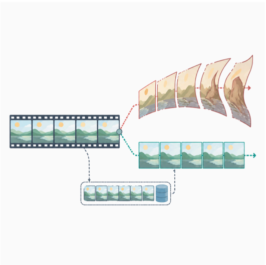
Diffusion · Autoregressive

C8 Evaluation — 25.5%

1. **Open-world quality**
   WorldScore · Nano-WM
2. **Game / driving benchmarks**
   WBench · driving suites
3. **Closed-loop embodied**
   World-in-World · RoboWM-Bench

Benchmark (18 papers)

C5 Stability STRENGTH — 40%



1. **Self-adversarial training**
   Self-Forcing · Matrix-Game2.0
2. **Memory (implicit/explicit)**
   VMem · Long-term Spatial Mem
3. **Causal consistency + theory**
   WorldCraft TASP · Fine-flow

Autoregressive · Diffusion